

Vibe Coding Python



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What Is Python?

- **Python** is a popular coding language
 - Think of it like an **execution engine** — it runs instructions on your machine.
- A Python file has the extension `.py`
 - Python files are just text files
 - They are repeatable scripts you can run
- You run a Python file to produce outputs (text, plots, images, etc.)
- Your vibe coder can handle all parts of the Python flow
 - Write Python files (for sure)
 - Run Python files (if you want)

```
# extract_leases.py – a reusable SOP
def main():
    leases = load_pdfs("sample_leases/")
    save_json(leases, "leases_data.json")
```



Python Packages

- Packages are bundles of code that have functions for specific purposes
- To use a package you need to do two things:
 - Install the package (do this once)
 - Import the package (do this in your python file)

Install once (terminal)

```
pip install pandas dash
```

Downloads code into your environment. Do this when setting up a project.

Import in code

```
import pandas as pd
```

Loads the package into *this* script so you can call its functions.

Some useful python packages:

`python-dotenv`

Load secrets from

`.env`

`openai`

/

`google-genai`

Call LLM APIs

`pydantic`

Validate schemas / types

`pandas`

Tables & analysis

`dash`

Interactive web apps

`plotly`

Charts & maps

API Keys & Security

- API — application programming interface
- Many applications need an **API key** — proof you can bill / access the service.
- Store keys as **environment variables** in a local `.env` file — **not in python files!**
- Load them in Python with the `python-dotenv` package.

```
# .env (never commit this file)
PORTKEY_API_KEY=dffds-...
GEMINI_API_KEY=f433d-...
STRIPE_API_KEY=jyut-...
ELEVENLABS_API_KEY=12de-...
```

- If a key leaks online, treat it as stolen and rotate it immediately
 - Public sites get scraped for credentials (sometimes [by AI](#))



Structured Data & `.json` Files

- JSON — JavaScript Object Notation
- JSON is a text format for structured data (key–value pairs, lists)
- AI models need a **strict schema** for inputs and outputs
 - Otherwise you get essays, not rows.
 - For complex agentic flows, we need JSON formats so data can be passed between agents safely

Messy text

"Tenant ACME Corp pays \$42/sf at 100 Chapel St, lease ends Dec 2027..."

Human-readable. Machine-hostile.



Clean JSON object

```
{  
  "tenant": "ACME",  
  "rent_psf": 42,  
  "address": "100 Chapel St",  
  "end": "2027-12-31"  
}
```

Key–value pairs.

Dashboards With Dash

- We can view the data through a web dashboard
- Python's Dash package lets us do this easily
 - Write the app in a file `app.py`
 - Run the file from the terminal

```
# app.py – a simple Dash app
from dash import Dash, html

app = Dash(__name__)
app.layout = html.Div([
    html.H1("Lease Dashboard"),
    html.P("Filter leases and explore the map."),
])

if __name__ == "__main__":
    app.run(debug=True)
```

```
# in the terminal, from your project folder
python app.py
# → open http://127.0.0.1:8050
```



Virtual Environments

- A **virtual environment** keeps a project self-contained
- It is a folder in your project with its own Python + installed packages
- Ask your vibe coder to create and use one for you
- Good software practice
 - Other people can recreate your project from `requirements.txt`
 - If a package version changes globally, your project is not affected

```
# create a virtual environment (once)
```

```
python -m venv .venv
```

```
# Windows: activate it
```

```
.venv\Scripts\activate
```

```
# Mac / Linux: activate it
```

```
source .venv/bin/activate
```

```
# then install packages into THIS project only
```

```
pip install -r requirements.txt
```

lec02/

app.py

requirements.txt

.env # secrets – gitignore this

.venv/ # created by venv

├── Scripts/ # Windows

│ ├── python.exe

│ ├── pip.exe

│ └── activate

├── bin/ # Mac / Linux

│ ├── python

│ ├── pip

│ └── activate

└── Lib/site-packages/

│ ├── dash/

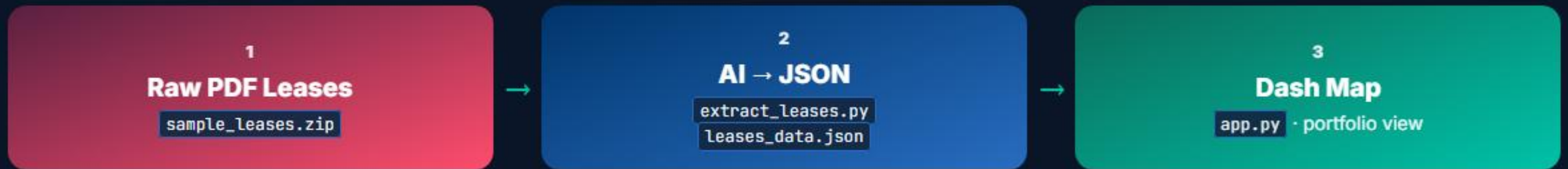
│ ├── plotly/

│ └── ...

Mapping Your Flow

- Before vibe coding, think first — then prompt
- Plan:
 - **Type** of project (Python script, Dash app, React, etc.)
 - Project **structure** and **flow**
 - **Data** needed (an API, or files already on your machine)
 - **Keys** you will need (`.env`)
 - **Packages** you will install
 - **Styles** you like (layout, colors, chart look)
- Draw a **flow diagram** so the AI builds the path you want

Example: Lease Analyzer



[Open the vibe coding page →](#)